

BOM — bom

Bill of Materials - V4b (VBF-T1R2-BOM-01) case t1_r2 energy ref 20.5534 Wh (cell/r4_v4b_energy.json:energy_wh)			
Component	Mass (g/cell)	Mass (kg/kWh)	Formula / source
Positive active material (NMC811)	15.399	0.7492	coating mass x 0.95; coating = layer_kg_m2.positive_electrode x 0.1027 m2; 95/3/2 wt% active/binder/conductive literature default (annotated estimate)
Positive binder (PVDF)	0.486	0.0237	coating mass x 0.03 (literature default, annotated)
Positive conductive additive (carbon black)	0.324	0.0158	coating mass x 0.02 (literature default, annotated)
Negative active material (graphite)	10.553	0.5135	coating mass x 0.95; coating = layer_kg_m2.negative_electrode x 0.1027 m2
Negative binder (CMC/SBR)	0.333	0.0162	coating mass x 0.03 (literature default, annotated)
Negative conductive additive	0.222	0.0108	coating mass x 0.02 (literature default, annotated)
Separator (polyolefin)	0.183	0.0089	layer_kg_m2.separator x 0.1027 m2
Positive current collector (Al)	2.218	0.1079	layer_kg_m2.positive_cc x 0.1027 m2 (8 um x 2700 kg/m3)
Negative current collector (Cu)	5.521	0.2686	layer_kg_m2.negative_cc x 0.1027 m2 (6 um x 8960 kg/m3)
Electrolyte (LiPF6/EC-EMC)	7.694	0.3743	pore volume 6.41 cm3 x 1.2 g/cm3 literature density (annotated estimate), fill factor 1.0
Enclosure / tabs	Not modeled	Not modeled	Not modeled (beyond pure simulation boundary)
Total (contract caliber, electrolyte excluded)	35.241	1.7146	
Total (incl. electrolyte estimate)	42.935	2.0889	